SyntHIR Github Documentation

Pavitra Chauhan

June 10, 2023

1 System Requirements/Pre-requites

1. Java version 17
2. Maven 4.0
3. Python 3.8

2 Components configuration and installation

2.1 Data Wrangling

It uses Azure Blob Storage for managing the CSV files uploaded and downloaded. To configure, add the following details of the Azure Blob Storage to the application.properties file:

1. azure.storage.blob.account-name
2. azure.storage.blob.account-key
3. azure.storage.blob.endpoint=http://xxxx.blob.core.windows.net
4. server.port=XXXX (Port on which the application will run)

2.2 FHIR Server Configuration

Please follow the following steps to configure 'Azure API for FHIR' as the FHIR server:

1. Create a subscription on the Azure portal
2. Create a resource, 'Azure API for FHIR', which is a part of the subscription in a resource group. A resource group is a container that holds and manages all those resources that need to be managed in a group. Link:Azure-Portal
3. Here is a resource called Azure Active Directory (AAD) which is used by Azure for identity and access management services. Each AAD can have single or multiple tenants, and each tenant has a Tenant ID associated with it. In AAD tenant, register a confidential client application. Each client application has an application client ID, and credentials. These credentials help the applications to identify themselves to authenticate when requesting access tokens. Create an application (App Registrations) in Azure Active Directory (AAD), and add it as an enterprise application in a tenant. These applications are registered in the AAD. Create the client secret ID and value in 'Certificates & secrets' of the created application, and make a note of the Client secret value as it will be used to call FHIR APIs. Link: Azure - AAD
4. Once the application is registered, it is granted access to the FHIR Server in the access control. There are various roles when granting access to the application such as FHIR Data Contributor, FHIR Data Converter, FHIR Data Exporter, FHIR Data Reader and FHIR Data Writer. Add the application created to the FHIR server using Role Assignment. Link: AZure-FHIR-Server

5. Note: So essentially, there is an FHIR server ('Azure API for FHIR') and an application is registered in it. FHIR APIs are accessible using the credentials of the application. Applications are registered in Azure Active Directory tenant. Each tenant can have multiple applications.

Configuration for the component It uses FHIR server for uploading the resources to the FHIR server. To configure, add following details of the FHIR server to the application.properties file: hdl.fhir.server.base.url = fhir server base Url hdl.azure.app.client.reg.id = clientid - Application client registration ID hdl.azure.app.client.reg.secret.value = clientsecret - Application client registration secret value server.port=80xx (Port on which the application will run) After the above steps, we are authorized to access the FHIR API. To access FHIR API, the following steps:

1. Since the FHIR service is secured by AAD, get an access token from the active directory. The request for getting an access token :
   (a) Request Type: POST
   (b) Request URL: https://login.microsoftonline.com/tenantid/oauth2/token (tenant ID- is the ID of the tenant registered under Azure Active Directory)
   (c) Body: In the 'x-www-form-urlencoded' format
2. We get the access token in response to the above request. Using the Azure AD access token, we can request FHIR resources. For example, to get a list of patients:
   (a) Request Type: GET
   (b) Request URL: fhirurl/Patient
   (c) Authorization: Set type as 'Bearer Token' and value as the token received from the Azure AD

```
{"menu": {
  "grant_type": "Client_Credentials",
  "client_id": "{{clientid}} (Is the client application ID)",
  "client_secret" : "{{clientsecret}} (Client secret credentials)",
  "resource" : "{{base_fhirurl}} (Base URL of the FHIR service)"
  "popup": {
}}
```

2.3 Gretel Configuration

3 API Documentation

3.1 Data Wrangling

3.1.1 Convert CSV to FHIR resources

1. Request URL: http://localhost:XXXX/api/v1/data-wrangling/convert/npr/csv-to-fhir
2. Request Type: POST
3. Request Body: Will have form-data. With CSV file in "file" request param.
4. Response Body: Will be a list of JSON objects (Example of single JSON object from the list)

```
[
    {
        "patient": {
            "resourceType": "Patient",
            "id": null,
```

```
                                "meta": null,
                                "identifier": [
                                    {
                                        "use": "temp",
                                        "system": null,
                                        "value": "34940"
                                    }
                                ],
                                "active": false,
                                "name": null,
                                "gender": "female",
                                "birthDate": "1941",
                                "deceasedBoolean": false,
                                "deceasedDateTime": "",
                                "address": [
                                    {
                                        "use": "",
                                        "city": "Hordaland Fylkeskommune",
                                        "district": "",
                                        "state": "",
                                        "postalCode": "12",
                                        "country": "Norway"
                                    }
                                ],
                                "extension": [
                                    {
                                        "valueString": "3"
                                    }
                                ]
                            },
                            "practitioner": {
                                "resourceType": "Practitioner",
                                "id": null,
                                "meta": null,
                                "identifier": [
                                    {
                                        "use": "temp",
                                        "system": null,
                                        "value": "44260"
                                    }
                                ],
                                "active": false,
                                "name": null,
                                "gender": "female",
                                "birthDate": "1954"
                            },
                            "location": {
                                "resourceType": "Location",
                                "id": null,
                                "meta": null,
                                "identifier": null,
                                "status": null,
                                "name": "Helse Bergen HF Haukeland",
                                "mode": null,
```

```
                                "address": null
                            },
                            "condition": {
                                "resourceType": "Condition",
                                "id": null,
                                "meta": null,
                                "identifier": null,
                                "category": null,
                                "code": {
                                    "coding": [
                                        {
                                            "code": "D11",
                                            "system": "http://hl7.org/fhir/sid/icd-10",
                                            "display": null
                                        }
                                    ],
                                    "text": "ICD-10 Codes"
                                },
                                "subject": {
                                    "reference": "",
                                    "identifier": {
                                        "use": null,
                                        "system": null,
                                        "value": ""
                                    },
                                    "display": "Patient associated with the condition",
                                    "type": "Patient"
                                },
                                "encounter": {
                                    "reference": "",
                                    "identifier": {
                                        "use": null,
                                        "system": null,
                                        "value": ""
                                    },
                                    "display": "Encounter associated with Patient",
                                    "type": "Encounter"
                                }
                            },
                            "encounter": {
                                "resourceType": "Encounter",
                                "id": null,
                                "meta": null,
                                "identifier": null,
                                "status": "finished",
                                "type": null,
                                "subject": {
                                    "reference": "",
                                    "identifier": {
                                        "use": null,
                                        "system": null,
                                        "value": ""
```

```
                    },
                    "display": "Patient Hospitalized",
                    "type": "Patient"
                },
                "location": [
                    {
                        "location": {
                            "reference": "",
                            "identifier": {
                                "use": null,
                                "system": null,
                                "value": ""
                            },
                            "display": "Institute Name where prescribed",
                            "type": "Location"
                        },
                        "status": null
                    }
                ],
                "hospitalization": {
                    "dischargeDisposition": {
                        "coding": [
                            {
                                "code": "Other",
                                "system": "http://terminology.hl7.org/CodeSystem/discharge-disposition",
                                "display": null
                            }
                        ],
                        "text": "Others"
                    }
                },
                "period": {
                    "start": "2012-12-07",
                    "end": "2012-12-07"
                },
                "diagnosis": null,
                "participant": [
                    {
                        "individual": {
                            "reference": "",
                            "identifier": {
                                "use": null,
                                "system": null,
                                "value": ""
                            },
                            "display": "Practitioner Details for the patient
```

```
                                                hospitalized",
                                            "type": "Practitioner"
                                        }
                                    }
                                ],
                                "class": {
                                    "code": "PRENC",
                                    "system": "http://terminology.hl7.org/CodeSystem/v3-ActCode",
                                    "display": "Patient arrival mode for the Encounter"
                                }
                            },
                            "medication": {
                                "resourceType": "Medication",
                                "id": null,
                                "meta": null,
                                "identifier": [
                                    {
                                        "use": null,
                                        "system": null,
                                        "value": "5390"
                                    }
                                ],
                                "code": {
                                    "coding": [
                                        {
                                            "code": "R03BB01",
                                            "system": "http://www.whocc.no/atc",
                                            "display": null
                                        }
                                    ],
                                    "text": "Atrovent inh aer 20 mcg/dose ff"
                                }
                            },
                            "medicationRequest": {
                                "resourceType": "MedicationRequest",
                                "id": null,
                                "meta": null,
                                "identifier": [
                                    {
                                        "use": null,
                                        "system": null,
                                        "value": "50442198"
                                    }
                                ],
                                "status": "unknown",
                                "intent": "option",
                                "category": [
                                    {
                                        "coding": [
                                            {
```

```
                                        "code": "3",
                                        "system": null,
                                        "display": null
                                    }
                                ],
                                "text": "
                                    Bl reseptordningen
                                        2, 3a, 3b, 4 og 5
                                    (gammel ordning      2,
                                     3, 4, 9, og 10a)"
                            }
                        ],
                        "medicationReference": {
                            "reference": "",
                            "identifier": {
                                "use": null,
                                "system": null,
                                "value": ""
                            },
                            "display": "Medications for
                               the prescription",
                            "type": "Medication"
                        },
                        "subject": {
                            "reference": "",
                            "identifier": {
                                "use": null,
                                "system": null,
                                "value": ""
                            },
                            "display": "Patient for the
                               prescription",
                            "type": "Patient"
                        },
                        "encounter": {
                            "reference": "",
                            "identifier": {
                                "use": null,
                                "system": null,
                                "value": ""
                            },
                            "display": "Encounter
                               associated with the
                               prescription",
                            "type": "Encounter"
                        },
                        "recorder": {
                            "reference": "",
                            "identifier": {
                                "use": null,
                                "system": null,
                                "value": ""
                            },
                            "display": "Practitioner who
                               prescribed the prescription
                               ",
```

```
                "type": "Practitioner"
            },
            "note": [
                {
                    "authorString": "Legal Reimbursement category for the prescription",
                    "text": "\   2 Forh ndsgodkjent refusjon (tidligere 9)"
                },
                {
                    "authorString": "Legal Reimbursement code for the prescription",
                    "text": "7"
                },
                {
                    "authorString": " Reimbursement code for the prescription - ICD/ICPC",
                    "text": "ICPC:R95"
                }
            ],
            "dosageInstruction": [
                {
                    "text": "Defined daily dose of the drug",
                    "doseAndRate": [
                        {
                            "doseQuantity": {
                                "value": 0.12,
                                "unit": "mg"
                            },
                            "rateQuantity": {
                                "value": 0.0,
                                "unit": "Per Day"
                            }
                        }
                    ]
                }
            ]
        },
        "medicationDispense": {
            "resourceType": " MedicationDispense",
            "id": null,
            "meta": null,
            "status": "unknown",
            "medicationReference": {
                "reference": "",
                "identifier": {
                    "use": null,
```

```
                                        "system": null,
                                        "value": ""
                                    },
                                    "display": "Medication details for the dispense",
                                    "type": "Medication"
                                },
                                "subject": {
                                    "reference": "",
                                    "identifier": {
                                        "use": null,
                                        "system": null,
                                        "value": ""
                                    },
                                    "display": "Patient for the prescription",
                                    "type": "Patient"
                                },
                                "authorizingPrescription": [
                                    {
                                        "reference": "",
                                        "identifier": {
                                            "use": null,
                                            "system": null,
                                            "value": ""
                                        },
                                        "display": "Prescription for the Medication",
                                        "type": "MedicationRequest"
                                    }
                                ],
                                "quantity": {
                                    "value": 3000.0,
                                    "unit": null
                                },
                                "daysSupply": {
                                    "value": 100000.0,
                                    "unit": null
                                },
                                "whenHandedOver": ""
                            }
                        }
                    ]
```


3.1.2 Convert FHIR resources to CSV

1. Request URL: httphttp://localhost:XXXX/api/v1/data-wrangling/convert/npr/fhir-to-csv
2. Request Type: POST
3. Request Body: Will be a list of JSON objects (Example of single JSON object from the list) (in JSON format)


```
                    [
                        {
                            "patient": {
```

```
                                "resourceType": "Patient",
                                "id": null,
                                "meta": null,
                                "identifier": [
                                    {
                                        "use": "temp",
                                        "system": null,
                                        "value": "34940"
                                    }
                                ],
                                "active": false,
                                "name": null,
                                "gender": "female",
                                "birthDate": "1941",
                                "deceasedBoolean": false,
                                "deceasedDateTime": "",
                                "address": [
                                    {
                                        "use": "",
                                        "city": "Hordaland
                                           Fylkeskommune",
                                        "district": "",
                                        "state": "",
                                        "postalCode": "12",
                                        "country": "Norway"
                                    }
                                ],
                                "extension": [
                                    {
                                        "valueString": "3"
                                    }
                                ]
                            },
                            "practitioner": {
                                "resourceType": "Practitioner",
                                "id": null,
                                "meta": null,
                                "identifier": [
                                    {
                                        "use": "temp",
                                        "system": null,
                                        "value": "44260"
                                    }
                                ],
                                "active": false,
                                "name": null,
                                "gender": "female",
                                "birthDate": "1954"
                            },
                            "location": {
                                "resourceType": "Location",
                                "id": null,
                                "meta": null,
                                "identifier": null,
                                "status": null,
                                "name": "Helse Bergen HF Haukeland
```

```
                                    ",
                                "mode": null,
                                "address": null
                            },
                            "condition": {
                                "resourceType": "Condition",
                                "id": null,
                                "meta": null,
                                "identifier": null,
                                "category": null,
                                "code": {
                                    "coding": [
                                        {
                                            "code": "D11",
                                            "system": "http://hl7.org/fhir/sid/icd-10",
                                            "display": null
                                        }
                                    ],
                                    "text": "ICD-10 Codes"
                                },
                                "subject": {
                                    "reference": "",
                                    "identifier": {
                                        "use": null,
                                        "system": null,
                                        "value": ""
                                    },
                                    "display": "Patient associated with the condition",
                                    "type": "Patient"
                                },
                                "encounter": {
                                    "reference": "",
                                    "identifier": {
                                        "use": null,
                                        "system": null,
                                        "value": ""
                                    },
                                    "display": "Encounter associated with Patient",
                                    "type": "Encounter"
                                }
                            },
                            "encounter": {
                                "resourceType": "Encounter",
                                "id": null,
                                "meta": null,
                                "identifier": null,
                                "status": "finished",
                                "type": null,
                                "subject": {
                                    "reference": "",
                                    "identifier": {
                                        "use": null,
```

```
                                        "system": null,
                                        "value": ""
                                    },
                                    "display": "Patient Hospitalized",
                                    "type": "Patient"
                                },
                                "location": [
                                    {
                                        "location": {
                                            "reference": "",
                                            "identifier": {
                                                "use": null,
                                                "system": null,
                                                "value": ""
                                            },
                                            "display": "Institute Name where prescribed",
                                            "type": "Location"
                                        },
                                        "status": null
                                    }
                                ],
                                "hospitalization": {
                                    "dischargeDisposition": {
                                        "coding": [
                                            {
                                                "code": "Other",
                                                "system": "http://terminology.hl7.org/CodeSystem/discharge-disposition",
                                                "display": null
                                            }
                                        ],
                                        "text": "Others"
                                    }
                                },
                                "period": {
                                    "start": "2012-12-07",
                                    "end": "2012-12-07"
                                },
                                "diagnosis": null,
                                "participant": [
                                    {
                                        "individual": {
                                            "reference": "",
                                            "identifier": {
                                                "use": null,
                                                "system": null,
                                                "value": ""
                                            },
                                            "display": "Practitioner
```

```
                                                Details for the patient hospitalized",
                                            "type": "Practitioner"
                                        }
                                    }
                                ],
                                "class": {
                                    "code": "PRENC",
                                    "system": "http://terminology.hl7.org/CodeSystem/v3-ActCode",
                                    "display": "Patient arrival mode for the Encounter"
                                }
                            },
                            "medication": {
                                "resourceType": "Medication",
                                "id": null,
                                "meta": null,
                                "identifier": [
                                    {
                                        "use": null,
                                        "system": null,
                                        "value": "5390"
                                    }
                                ],
                                "code": {
                                    "coding": [
                                        {
                                            "code": "R03BB01",
                                            "system": "http://www.whocc.no/atc",
                                            "display": null
                                        }
                                    ],
                                    "text": "Atrovent inh aer 20 mcg/dose ff"
                                }
                            },
                            "medicationRequest": {
                                "resourceType": "MedicationRequest",
                                "id": null,
                                "meta": null,
                                "identifier": [
                                    {
                                        "use": null,
                                        "system": null,
                                        "value": "50442198"
                                    }
                                ],
                                "status": "unknown",
                                "intent": "option",
                                "category": [
                                    {
```

```
                                        "coding": [
                                            {
                                                "code": "3",
                                                "system": null,
                                                "display": null
                                            }
                                        ],
                                        "text": "
                                            Bl reseptordningen
                                                2, 3a, 3b, 4 og 5
                                            (gammel ordning      2,
                                             3, 4, 9, og 10a)"
                                    }
                                ],
                                "medicationReference": {
                                    "reference": "",
                                    "identifier": {
                                        "use": null,
                                        "system": null,
                                        "value": ""
                                    },
                                    "display": "Medications for
                                        the prescription",
                                    "type": "Medication"
                                },
                                "subject": {
                                    "reference": "",
                                    "identifier": {
                                        "use": null,
                                        "system": null,
                                        "value": ""
                                    },
                                    "display": "Patient for the
                                        prescription",
                                    "type": "Patient"
                                },
                                "encounter": {
                                    "reference": "",
                                    "identifier": {
                                        "use": null,
                                        "system": null,
                                        "value": ""
                                    },
                                    "display": "Encounter
                                        associated with the
                                        prescription",
                                    "type": "Encounter"
                                },
                                "recorder": {
                                    "reference": "",
                                    "identifier": {
                                        "use": null,
                                        "system": null,
                                        "value": ""
                                    },
                                    "display": "Practitioner who
```

```
                                        prescribed the prescription",
                                    "type": "Practitioner"
                                },
                                "note": [
                                    {
                                        "authorString": "Legal Reimbursement category for the prescription",
                                        "text": "   2 Forh ndsgodkjent refusjon (tidligere 9)"
                                    },
                                    {
                                        "authorString": "Legal Reimbursement code for the prescription",
                                        "text": "7"
                                    },
                                    {
                                        "authorString": "Reimbursement code for the prescription - ICD/ICPC",
                                        "text": "ICPC:R95"
                                    }
                                ],
                                "dosageInstruction": [
                                    {
                                        "text": "Defined daily dose of the drug",
                                        "doseAndRate": [
                                            {
                                                "doseQuantity": {
                                                    "value": 0.12,
                                                    "unit": "mg"
                                                },
                                                "rateQuantity": {
                                                    "value": 0.0,
                                                    "unit": "Per Day"
                                                }
                                            }
                                        ]
                                    }
                                ]
                            },
                            "medicationDispense": {
                                "resourceType": "MedicationDispense",
                                "id": null,
                                "meta": null,
                                "status": "unknown",
                                "medicationReference": {
                                    "reference": "",
```

```
                                    "identifier": {
                                        "use": null,
                                        "system": null,
                                        "value": ""
                                    },
                                    "display": "Medication details for the dispense",
                                    "type": "Medication"
                                },
                                "subject": {
                                    "reference": "",
                                    "identifier": {
                                        "use": null,
                                        "system": null,
                                        "value": ""
                                    },
                                    "display": "Patient for the prescription",
                                    "type": "Patient"
                                },
                                "authorizingPrescription": [
                                    {
                                        "reference": "",
                                        "identifier": {
                                            "use": null,
                                            "system": null,
                                            "value": ""
                                        },
                                        "display": "Prescription for the Medication",
                                        "type": "MedicationRequest"
                                    }
                                ],
                                "quantity": {
                                    "value": 3000.0,
                                    "unit": null
                                },
                                "daysSupply": {
                                    "value": 100000.0,
                                    "unit": null
                                },
                                "whenHandedOver": ""
                            }
                        }
                    ]
```


4. Response Body: CSV file

3.2 FHIR Adapter

3.2.1 Upload FHIR resources to FHIR server

1. Request URL: http://localhost:XXXX/api/v1/fhir-server/upload?fhirServerUrl=https://hdlsyntheticdata.azureheal
2. Request Type: POST
3. Request Body: Will be a list of JSON objects (Example of single JSON object from the list)

```
[
    {
        "patient": {
            "resourceType": "Patient",
            "id": null,
            "meta": null,
            "identifier": [
                {
                    "use": "temp",
                    "system": null,
                    "value": "34940"
                }
            ],
            "active": false,
            "name": null,
            "gender": "female",
            "birthDate": "1941",
            "deceasedBoolean": false,
            "deceasedDateTime": "",
            "address": [
                {
                    "use": "",
                    "city": "Hordaland
                       Fylkeskommune",
                    "district": "",
                    "state": "",
                    "postalCode": "12",
                    "country": "Norway"
                }
            ],
            "extension": [
                {
                    "valueString": "3"
                }
            ]
        },
        "practitioner": {
            "resourceType": "Practitioner",
            "id": null,
            "meta": null,
            "identifier": [
                {
                    "use": "temp",
                    "system": null,
                    "value": "44260"
                }
            ],
            "active": false,
            "name": null,
            "gender": "female",
            "birthDate": "1954"
        },
        "location": {
            "resourceType": "Location",
            "id": null,
            "meta": null,
```

```
                    "identifier": null,
                    "status": null,
                    "name": "Helse Bergen HF Haukeland",
                    "mode": null,
                    "address": null
                },
                "condition": {
                    "resourceType": "Condition",
                    "id": null,
                    "meta": null,
                    "identifier": null,
                    "category": null,
                    "code": {
                        "coding": [
                            {
                                "code": "D11",
                                "system": "http://hl7.org/fhir/sid/icd-10",
                                "display": null
                            }
                        ],
                        "text": "ICD-10 Codes"
                    },
                    "subject": {
                        "reference": "",
                        "identifier": {
                            "use": null,
                            "system": null,
                            "value": ""
                        },
                        "display": "Patient associated with the condition",
                        "type": "Patient"
                    },
                    "encounter": {
                        "reference": "",
                        "identifier": {
                            "use": null,
                            "system": null,
                            "value": ""
                        },
                        "display": "Encounter associated with Patient",
                        "type": "Encounter"
                    }
                },
                "encounter": {
                    "resourceType": "Encounter",
                    "id": null,
                    "meta": null,
                    "identifier": null,
                    "status": "finished",
                    "type": null,
                    "subject": {
                        "reference": "",
                        "identifier": {
```

```
                                "use": null,
                                "system": null,
                                "value": ""
                            },
                            "display": "Patient Hospitalized",
                            "type": "Patient"
                        },
                        "location": [
                            {
                                "location": {
                                    "reference": "",
                                    "identifier": {
                                        "use": null,
                                        "system": null,
                                        "value": ""
                                    },
                                    "display": "Institute Name where prescribed",
                                    "type": "Location"
                                },
                                "status": null
                            }
                        ],
                        "hospitalization": {
                            "dischargeDisposition": {
                                "coding": [
                                    {
                                        "code": "Other",
                                        "system": "http://terminology.hl7.org/CodeSystem/discharge-disposition",
                                        "display": null
                                    }
                                ],
                                "text": "Others"
                            }
                        },
                        "period": {
                            "start": "2012-12-07",
                            "end": "2012-12-07"
                        },
                        "diagnosis": null,
                        "participant": [
                            {
                                "individual": {
                                    "reference": "",
                                    "identifier": {
                                        "use": null,
                                        "system": null,
                                        "value": ""
                                    },
                                    "display": "Practitioner Details for the patient hospitalized",
```

```
                                        "type": "Practitioner"
                                    }
                                }
                            ],
                            "class": {
                                "code": "PRENC",
                                "system": "http://terminology.hl7.org/CodeSystem/v3-ActCode",
                                "display": "Patient arrival mode for the Encounter"
                            }
                        },
                        "medication": {
                            "resourceType": "Medication",
                            "id": null,
                            "meta": null,
                            "identifier": [
                                {
                                    "use": null,
                                    "system": null,
                                    "value": "5390"
                                }
                            ],
                            "code": {
                                "coding": [
                                    {
                                        "code": "R03BB01",
                                        "system": "http://www.whocc.no/atc",
                                        "display": null
                                    }
                                ],
                                "text": "Atrovent inh aer 20mcg/dose ff"
                            }
                        },
                        "medicationRequest": {
                            "resourceType": "MedicationRequest",
                            "id": null,
                            "meta": null,
                            "identifier": [
                                {
                                    "use": null,
                                    "system": null,
                                    "value": "50442198"
                                }
                            ],
                            "status": "unknown",
                            "intent": "option",
                            "category": [
                                {
                                    "coding": [
                                        {
                                            "code": "3",
                                            "system": null,
                                            "display": null
```

```
                                        }
                                    ],
                                    "text": "Bl reseptordningen 2, 3a, 3b, 4 og 5 (gammel ordning 2, 3, 4, 9, og 10a)"
                                }
                            ],
                            "medicationReference": {
                                "reference": "",
                                "identifier": {
                                    "use": null,
                                    "system": null,
                                    "value": ""
                                },
                                "display": "Medications for the prescription",
                                "type": "Medication"
                            },
                            "subject": {
                                "reference": "",
                                "identifier": {
                                    "use": null,
                                    "system": null,
                                    "value": ""
                                },
                                "display": "Patient for the prescription",
                                "type": "Patient"
                            },
                            "encounter": {
                                "reference": "",
                                "identifier": {
                                    "use": null,
                                    "system": null,
                                    "value": ""
                                },
                                "display": "Encounter associated with the prescription",
                                "type": "Encounter"
                            },
                            "recorder": {
                                "reference": "",
                                "identifier": {
                                    "use": null,
                                    "system": null,
                                    "value": ""
                                },
                                "display": "Practitioner who prescribed the prescription",
                                "type": "Practitioner"
                            },
                            "note": [
                                {
                                    "authorString": "Legal Reimbursement category for
```

```
                                the prescription",
                            "text": "   2
                                Forh ndsgodkjent refusjon
                                (tidligere    9)"
                        },
                        {
                            "authorString": "Legal
                                Reimbursement code for the
                                prescription",
                            "text": "7"
                        },
                        {
                            "authorString": "Reimbursement
                                 code for the prescription
                                - ICD/ICPC",
                            "text": "ICPC:R95"
                        }
                    ],
                    "dosageInstruction": [
                        {
                            "text": "Defined daily dose of
                                 the drug",
                            "doseAndRate": [
                                {
                                    "doseQuantity": {
                                        "value": 0.12,
                                        "unit": "mg"
                                    },
                                    "rateQuantity": {
                                        "value": 0.0,
                                        "unit": "Per Day"
                                    }
                                }
                            ]
                        }
                    ]
                },
                "medicationDispense": {
                    "resourceType": "MedicationDispense",
                    "id": null,
                    "meta": null,
                    "status": "unknown",
                    "medicationReference": {
                        "reference": "",
                        "identifier": {
                            "use": null,
                            "system": null,
                            "value": ""
                        },
                        "display": "Medication details for
                            the dispense",
                        "type": "Medication"
                    },
                    "subject": {
                        "reference": "",
                        "identifier": {
```

```
                                        "use": null,
                                        "system": null,
                                        "value": ""
                                    },
                                    "display": "Patient for the prescription",
                                    "type": "Patient"
                                },
                                "authorizingPrescription": [
                                    {
                                        "reference": "",
                                        "identifier": {
                                            "use": null,
                                            "system": null,
                                            "value": ""
                                        },
                                        "display": "Prescription for the Medication",
                                        "type": "MedicationRequest"
                                    }
                                ],
                                "quantity": {
                                    "value": 3000.0,
                                    "unit": null
                                },
                                "daysSupply": {
                                    "value": 100000.0,
                                    "unit": null
                                },
                                "whenHandedOver": ""
                            }
                        }
                    ]
```


3.2.2 Download FHIR resources from FHIR server

1. Request URL: http://localhost:XXXX/api/v1/fhir-server/download?fhirServerUrl=https://hdlsyntheticdata.azureh
2. Request Type: GET
3. Response Body: Will be a list of JSON objects (Example of single JSON object from the list)


```
[
    {
        "patient": {
            "resourceType": "Patient",
            "id": "9dbcfce2-2c3a-476a-9b39-eead46d3c725",
            "meta": {
                "profile": null,
                "versionId": "1",
                "lastUpdated": "2023-06-05T22:03:13.987+00:00"
            },
            "identifier": [
                {
                    "use": "temp",
```

```
                                        "system": null,
                                        "value": "34940"
                                    }
                                ],
                                "active": false,
                                "name": null,
                                "gender": "female",
                                "birthDate": "1941",
                                "deceasedBoolean": false,
                                "deceasedDateTime": null,
                                "address": [
                                    {
                                        "use": null,
                                        "city": "Hordaland Fylkeskommune",
                                        "district": null,
                                        "state": null,
                                        "postalCode": "12",
                                        "country": "Norway"
                                    }
                                ],
                                "extension": [
                                    {
                                        "valueString": "3"
                                    }
                                ]
                            },
                            "practitioner": {
                                "resourceType": "Practitioner",
                                "id": "6f8f2364-e104-47e6-a98d-3614e9d5337d",
                                "meta": {
                                    "profile": null,
                                    "versionId": "1",
                                    "lastUpdated": "2023-06-05T22:03:14.824+00:00"
                                },
                                "identifier": [
                                    {
                                        "use": "temp",
                                        "system": null,
                                        "value": "44260"
                                    }
                                ],
                                "active": false,
                                "name": null,
                                "gender": "female",
                                "birthDate": "1954"
                            },
                            "location": {
                                "resourceType": "Location",
                                "id": "e1318990-70da-46ba-bc24-508b4c8b5332",
                                "meta": {
                                    "profile": null,
                                    "versionId": "1",
```

```
                                "lastUpdated": "2023-06-05T22:03:15.288+00:00"
                            },
                            "identifier": null,
                            "status": null,
                            "name": "Helse Bergen HF Haukeland",
                            "mode": null,
                            "address": null
                        },
                        "condition": {
                            "resourceType": "Condition",
                            "id": "dd1641fb-2c01-4a92-85f8-61bfe6ef6bbe",
                            "meta": {
                                "profile": null,
                                "versionId": "1",
                                "lastUpdated": "2023-06-05T22:03:15.763+00:00"
                            },
                            "identifier": null,
                            "category": null,
                            "code": {
                                "coding": [
                                    {
                                        "code": "D11",
                                        "system": "http://hl7.org/fhir/sid/icd-10",
                                        "display": null
                                    }
                                ],
                                "text": "ICD-10 Codes"
                            },
                            "subject": {
                                "reference": "Patient/9dbcfce2-2c3a-476a-9b39-eead46d3c725",
                                "identifier": {
                                    "use": null,
                                    "system": null,
                                    "value": "9dbcfce2-2c3a-476a-9b39-eead46d3c725"
                                },
                                "display": "Patient associated with the condition",
                                "type": "Patient"
                            },
                            "encounter": {
                                "reference": "Encounter/0e5b5fdd-b3a1-40cb-b8eb-e17c6c97c41d",
                                "identifier": {
                                    "use": null,
                                    "system": null,
                                    "value": "0e5b5fdd-b3a1-40
```

```
                                        cb-b8eb-e17c6c97c41d"
                                    },
                                    "display": "Encounter associated with Patient",
                                    "type": "Encounter"
                                }
                            },
                            "encounter": {
                                "resourceType": "Encounter",
                                "id": "0e5b5fdd-b3a1-40cb-b8eb-e17c6c97c41d",
                                "meta": {
                                    "profile": null,
                                    "versionId": "1",
                                    "lastUpdated": "2023-06-05T22:03:15.506+00:00"
                                },
                                "extension": null,
                                "identifier": null,
                                "status": "finished",
                                "type": null,
                                "subject": {
                                    "reference": "Patient/9dbcfce2-2c3a-476a-9b39-eead46d3c725",
                                    "identifier": {
                                        "use": null,
                                        "system": null,
                                        "value": "9dbcfce2-2c3a-476a-9b39-eead46d3c725"
                                    },
                                    "display": "Patient Hospitalized",
                                    "type": "Patient"
                                },
                                "location": [
                                    {
                                        "location": {
                                            "reference": "Location/e1318990-70da-46ba-bc24-508b4c8b5332",
                                            "identifier": {
                                                "use": null,
                                                "system": null,
                                                "value": "e1318990-70da-46ba-bc24-508b4c8b5332"
                                            },
                                            "display": "Institute Name where prescribed",
                                            "type": "Location"
                                        },
                                        "status": null
                                    }
```

```
                    ],
                    "hospitalization": {
                        "dischargeDisposition": {
                            "coding": [
                                {
                                    "code": "Other",
                                    "system": "http://terminology.hl7.org/CodeSystem/discharge-disposition",
                                    "display": null
                                }
                            ],
                            "text": "Others"
                        }
                    },
                    "period": {
                        "start": "2012-12-07",
                        "end": "2012-12-07"
                    },
                    "participant": [
                        {
                            "individual": {
                                "reference": "Practitioner/6f8f2364-e104-47e6-a98d-3614e9d5337d",
                                "identifier": {
                                    "use": null,
                                    "system": null,
                                    "value": "6f8f2364-e104-47e6-a98d-3614e9d5337d"
                                },
                                "display": "Practitioner Details for the patient hospitalized",
                                "type": "Practitioner"
                            }
                        }
                    ],
                    "class": {
                        "code": "PRENC",
                        "system": "http://terminology.hl7.org/CodeSystem/v3-ActCode",
                        "display": "Patient arrival mode for the Encounter"
                    }
                },
                "medication": {
                    "resourceType": "Medication",
                    "id": "1de886f3-2471-4825-9e2a-193
```

```
1580fc2d0",
"meta": {
    "profile": null,
    "versionId": "1",
    "lastUpdated": "2023-06-05T22:03:15.993+00:00"
},
"identifier": [
    {
        "use": null,
        "system": null,
        "value": "5390"
    }
],
"code": {
    "coding": [
        {
            "code": "R03BB01",
            "system": "http://www.whocc.no/atc",
            "display": null
        }
    ],
    "text": "Atrovent inh aer 20 mcg/dose ff"
}
},
"medicationRequest": {
    "resourceType": "MedicationRequest",
    "id": "18be387f-5343-4fba-9890-2f61101280c6",
    "meta": {
        "profile": null,
        "versionId": "1",
        "lastUpdated": "2023-06-05T22:03:16.184+00:00"
    },
    "identifier": [
        {
            "use": null,
            "system": null,
            "value": "50442198"
        }
    ],
    "status": "unknown",
    "intent": "option",
    "category": [
        {
            "coding": [
                {
                    "code": "3",
                    "system": null,
                    "display": null
                }
            ],
```

```
                "text": "Bl reseptordningen 2, 3a, 3b, 4 og 5 (gammel ordning 2, 3, 4, 9, og 10a)"
            }
        ],
        "medicationReference": {
            "reference": "Medication/1de886f3-2471-4825-9e2a-1931580fc2d0",
            "identifier": {
                "use": null,
                "system": null,
                "value": "1de886f3-2471-4825-9e2a-1931580fc2d0"
            },
            "display": "Medications for the prescription",
            "type": "Medication"
        },
        "subject": {
            "reference": "Patient/9dbcfce2-2c3a-476a-9b39-eead46d3c725",
            "identifier": {
                "use": null,
                "system": null,
                "value": "9dbcfce2-2c3a-476a-9b39-eead46d3c725"
            },
            "display": "Patient for the prescription",
            "type": "Patient"
        },
        "encounter": {
            "reference": "Encounter/0e5b5fdd-b3a1-40cb-b8eb-e17c6c97c41d",
            "identifier": {
                "use": null,
                "system": null,
                "value": "0e5b5fdd-b3a1-40cb-b8eb-e17c6c97c41d"
            },
            "display": "Encounter associated with the prescription",
            "type": "Encounter"
        },
        "recorder": {
            "reference": "Practitioner/6f8f2364-e104-47e6-a98d-3614e9d5337d",
            "identifier": {
                "use": null,
```

```
                                        "system": null,
                                        "value": "6f8f2364-e104-47e6-a98d-3614e9d5337d"
                                    },
                                    "display": "Practitioner who prescribed the prescription",
                                    "type": "Practitioner"
                                },
                                "note": [
                                    {
                                        "authorString": "Legal Reimbursement category for the prescription",
                                        "text": "\   2 Forh ndsgodkjent refusjon (tidligere \ 9)"
                                    },
                                    {
                                        "authorString": "Legal Reimbursement code for the prescription",
                                        "text": "7"
                                    },
                                    {
                                        "authorString": "Reimbursement code for the prescription - ICD/ICPC",
                                        "text": "ICPC:R95"
                                    }
                                ],
                                "dosageInstruction": [
                                    {
                                        "text": "Defined daily dose of the drug",
                                        "doseAndRate": [
                                            {
                                                "doseQuantity": {
                                                    "value": 0.12,
                                                    "unit": "mg"
                                                },
                                                "rateQuantity": {
                                                    "value": 0.0,
                                                    "unit": "Per Day"
                                                }
                                            }
                                        ]
                                    }
                                ]
                            },
                            "medicationDispense": {
                                "resourceType": "MedicationDispense",
```

```
                "id": "5c5ff711-0cf7-469d-a9e2-4baf8f6cc16f",
                "meta": {
                    "profile": null,
                    "versionId": "1",
                    "lastUpdated": "2023-06-05T22:03:16.319+00:00"
                },
                "status": "unknown",
                "medicationReference": {
                    "reference": "Medication/1de886f3-2471-4825-9e2a-1931580fc2d0",
                    "identifier": {
                        "use": null,
                        "system": null,
                        "value": "1de886f3-2471-4825-9e2a-1931580fc2d0"
                    },
                    "display": "Medication details for the dispense",
                    "type": "Medication"
                },
                "subject": {
                    "reference": "Patient/9dbcfce2-2c3a-476a-9b39-eead46d3c725",
                    "identifier": {
                        "use": null,
                        "system": null,
                        "value": "9dbcfce2-2c3a-476a-9b39-eead46d3c725"
                    },
                    "display": "Patient for the prescription",
                    "type": "Patient"
                },
                "authorizingPrescription": [
                    {
                        "reference": "MedicationRequest/18be387f-5343-4fba-9890-2f61101280c6",
                        "identifier": {
                            "use": null,
                            "system": null,
                            "value": "18be387f-5343-4fba-9890-2f61101280c6"
                        },
                        "display": "Prescription for the Medication",
                        "type": "MedicationRequest"
                    }
                ],
```

```
                                    "quantity": {
                                        "value": 3000.0,
                                        "unit": null
                                    },
                                    "daysSupply": {
                                        "value": 100000.0,
                                        "unit": null
                                    },
                                    "whenHandedOver": null
                                }
                            }
                        ]
```


3.3 Synthetic FHIR Data Generator

3.3.1 Generate Synthetic Data and download it

1. Request URL: http://localhost:8080/api/v1/synthetic/generate-data?numberOfSynRecords=10000
2. Request Type: POST
3. Request Body: Will have form data. With CSV file in "file" key and numberOfSynRecords as the query parameter. (Upload a file against which synthetic data is to be generated)
4. Response Body: Zip file with synthetic data

3.4 SyntHIR Pipeline

3.4.1 Convert CSV to FHIR and upload to Sensitive FHIR server

3.4.2 Download FHIR resources from Sensitive FHIR server and convert to CSV

3.4.3 Generate Synthetic data, convert to FHIR resources and upload to Synthetic FHIR server

4 Components Deployment

Each of the components is developed using the Spring boot framework, which has an embedded tomcat server. Following the following steps to deploy and run each of the components independently:

1. Set the port number of each of the components in the application configuration file
2. Build each component with the command: 'mvn clean install'.
3. To run each component: java -jar component-name.0.0.1.jar